Lab 3: Identify the Classes

Draw the Class Diagram

Digital Document Verification System

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1 Introduction

The **Digital Document Verification System** is a web-based platform designed to verify the authenticity and validity of digital documents such as certificates, identity documents, academic records, licenses, and official letters. This lab focuses on identifying the **classes** from the Software Requirements Specification (SRS) and Use Case specifications, and representing them in a **Class Diagram**.

A class is a blueprint for creating objects. It encapsulates **attributes** (data) and **methods** (behavior). Identifying classes is a crucial step in object-oriented analysis and design, as it forms the foundation for the system's architecture.

2 Class Identification

Classes were identified from the SRS document, Use Case specifications, and the system's functional requirements. The following categories of classes were considered:

- **Entity Classes**: Represent persistent data (e.g., User, Document).
- **Boundary Classes**: Represent interfaces between actors and the system (e.g., Report, Notification).
- **Control Classes**: Represent business logic and coordination (e.g., VerificationRecord, DigitalSignature).

2.1 Identified Classes

S.No	Class Name	Description
1	User	Represents a system user with authentication credentials and role.
2	EndUser	A specialized user who uploads and verifies documents.
3	Verifier	A specialized user authorized to verify documents on behalf of an organization.
4	Administrator	A specialized user with full system privileges.
5	Document	Represents a digital document uploaded for verification.
6	VerificationRecord	Stores the result and timestamp of a verification activity.
7	QRCode	Represents a QR code extracted from a document.
8	DigitalSignature	Represents a digital signature associated with a document.
9	Report	Represents a generated verification report.
10	Notification	Represents a notification sent to a user.

S.No	Class Name	Description
11	AuditLog	Records all user activities and system events.
12	IssuingAuthority	Represents an organization that issues and maintains document records.
13	DocumentRecord	Represents the official record of a document maintained by an issuing authority.
14	VerificationRule	Represents configurable rules used during verification.

3 Class Details

3.1 User (Abstract Class)

Attributes:

- - userId: int
- - name: String
- - email: String
- - password: String
- - role: String

Methods:

- + login()
- + logout()
- + register()

3.2 EndUser (Subclass of User)

Attributes:

- - documentsUploaded: int

Methods:

- + uploadDocument()
- + viewResult()

3.3 Verifier (Subclass of User)

Attributes:

- - organization: String
- - permissions: List

Methods:

- + verifyDocument()
- + viewDetails()

3.4 Administrator (Subclass of User)

Attributes:

- - privilegeLevel: int

Methods:

- + manageUsers()

- + reviewLogs()

3.5 Document

Attributes:
- - documentId: int
- - fileName: String
- - filePath: String
- - uploadDate: Date
- - status: String

Methods:
- + upload()
- + process()
- + getDetails()

3.6 VerificationRecord

Attributes:
- - recordId: int
- - documentId: int
- - result: String
- - timestamp: DateTime

Methods:
- + save()
- + getResult()

3.7 QRCode

Attributes:
- - qrId: int
- - encodedData: String
- - isValid: boolean

Methods:
- + scan()
- + validate()

3.8 DigitalSignature

Attributes:
- - signatureId: int
- - certificate: String
- - issuer: String
- - isValid: boolean

Methods:
- + verify()
- + detectTampering()

3.9 Report

Attributes:

- - reportId: int
- - recordId: int
- - filePath: String
- - generatedDate: Date

Methods:

- + generate()
- + download()

3.10 Notification

Attributes:

- - notifId: int
- - message: String
- - type: String
- - status: String

Methods:

- + send()
- + markRead()

3.11 AuditLog

Attributes:

- - logId: int
- - userId: int
- - action: String
- - timestamp: DateTime

Methods:

- + record()
- + display()

3.12 IssuingAuthority

Attributes:

- - authorityId: int
- - name: String
- - apiEndpoint: String

Methods:

- + provideRecord()
- + updateStatus()

3.13 DocumentRecord

Attributes:

- - recordId: int
- - documentId: int
- - issuerId: int
- - issueDate: Date

Methods:

- + fetch()
- + compare()

3.14 VerificationRule

Attributes:

- - ruleId: int
- - ruleName: String
- - condition: String
- - isActive: boolean

Methods:

- + apply()
- + update()

4 Relationships Between Classes

S.No	Class 1	Class 2	Relationship	Description
1	User	EndUser	Generalization	EndUser inherits from User.
2	User	Verifier	Generalization	Verifier inherits from User.
3	User	Administrator	Generalization	Administrator inherits from User.
4	EndUser	Document	Association	EndUser uploads Document.
5	Verifier	Document	Association	Verifier verifies Document.
6	Document	VerificationRecord	Association	Document has VerificationRecord.
7	VerificationRecord	QRCode	Association	VerificationRecord includes QRCode.
8	VerificationRecord	DigitalSignature	Association	VerificationRecord includes DigitalSignature.
9	VerificationRecord	Report	Association	VerificationRecord generates Report.
10	User	Notification	Association	User receives Notification.
11	User	AuditLog	Association	User actions recorded in AuditLog.
12	IssuingAuthority	DocumentRecord	Association	IssuingAuthority maintains DocumentRecord.

S.No	Class 1	Class 2	Relationship	Description
13	DocumentRecord	Document	Association	DocumentRecord corresponds to Document.
14	Administrator	VerificationRule	Association	Administrator configures VerificationRule.
15	VerificationRule	VerificationRecord	Association	VerificationRule applied to VerificationRecord.

5 Class Diagram

The class diagram below represents the static structure of the Digital Document Verification System. It shows the classes, their attributes, methods, and the relationships between them.

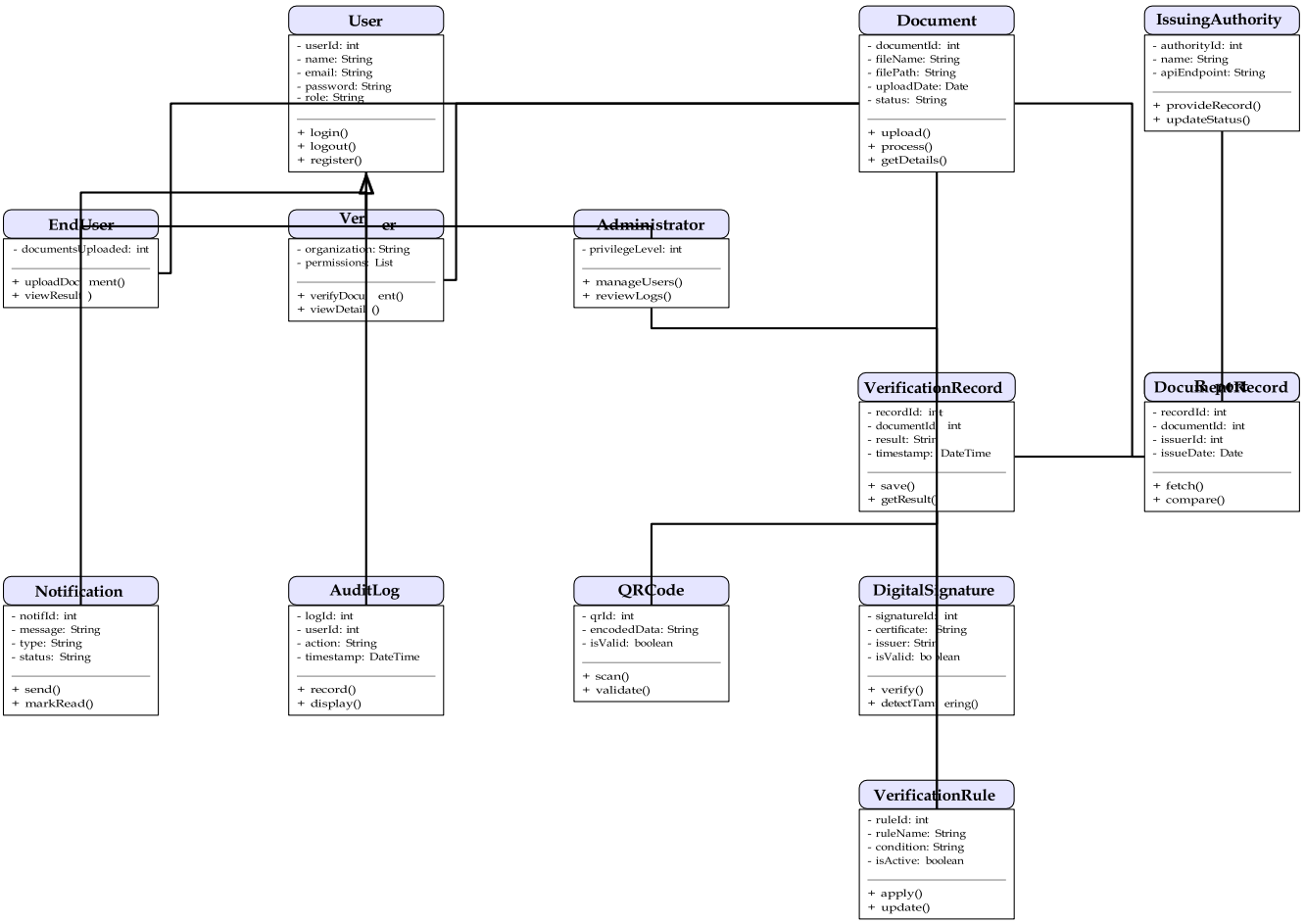


Figure 1: Class Diagram — Digital Document Verification System

6 Conclusion

In this lab, we successfully identified **14 classes** for the Digital Document Verification System and represented them in a class diagram. The classes were derived from the SRS document and Use Case specifications. The class diagram captures the static structure of the system, showing attributes, methods, and relationships such as **generalization** and **association**. This model serves as a foundation for the subsequent design and implementation phases.

References

[1] Software Requirements Specification (SRS) — Digital Document Verification System, Version 1.0.

[2] Use Case Diagram and Specifications — Digital Document Verification System.

[3] IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.

[4] ISO/IEC 27001:2013 — Information Security Management Systems.